

FPGAs without even writing low-level VHDL code!

Meeting the needs of the wireless world

The emergence and popularity of new, high-speed wireless technologies is greatly influencing changes in the test industry.

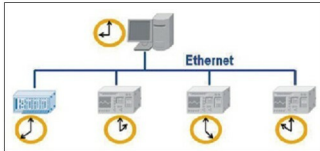
The high speed requirement of wireless devices and applications has resulted in wireless networks with more spectrum, greater data throughput, technologies such as orthogonal frequency-division multiple access, and advanced antennae with technologies like multiple-input multiple-output and beam-forming. Testing plays a huge role in making all these new technologies work together.

Today's wireless test equipment have to cover higher bandwidths and many parts of the spectrum. The solutions need to look beyond the channel frequencies specified for the DUT to help detect and minimise RF interference. They also need to take into consideration the security requirements of wireless devices, not to forget their physical robustness and even weather-resistance.

What is more, since the high-speed wireless industry is rapidly growing with new devices and protocols emerging everyday, the test solutions need to be very flexible and customisable. Hence these solutions often comprise standards-based software that can be downloaded to hardware instrumentation and modular platforms—to quickly put together a test system that meets the specific requirements of the project. The solutions must also be simple and foolproof because these might be used by design and field repair engineers with relatively little experience. Plus, since many wireless technologies are meant for low-cost devices, the test solutions must enable manufacturers to verify performance without increasing end-system cost.

Reducing time and costs

"The primary focus of test instrument manufacturers is to accurately capture



IEEE 1588's precision time protocol makes it possible to synchronise distributed clocks with an accuracy of less than one microsecond via Ethernet

a signal or create a stimulus. However, demands for lower costs equally drive these specifications in order to shorten tests and hence lower its cost," says Stasonis.

Indeed, in today's dynamic marketplace there is a great pressure on manufacturers to introduce new models at short intervals. Consequently, there is a pressure on test engineers to come up with solutions that reduce the testing time and cost while improving the effectiveness of the testing process. This is often achieved by test systems that feature faster signal capture, faster data transfer to the PC for analysis, and more intelligent drivers to shorten development times and minimise test steps.

Cost is as important as speed, because the cost of the test process influences the cost of the device as well. The emergence of flexible test systems that use smart software, modular hardware, FPGAs, etc has had a significant impact on the cost of the test process, as the same solution can be used to test multiple devices.

Another technology that has gained popularity in this context is the built-in self-test (BIST), where a device or machinery is also empowered with the ability to test itself.

"Test departments are constantly under pressure to reduce costs, and with less and less test-point access, BIST can be critical for efficient test," says Stasonis. Despite the reduced cost and time for testing, BIST is very reliable and is often used in critical spaces such as avionics and defence.

Sagar Patankar, senior program

manager, KPIT Cummins, says that test accelerators used in software testing also help meet these time and cost requirements. "Test accelerators are built with three goals: high quality, reduce time-to-mar-

ket and cost savings," he says. Some of KPIT Cummins' innovations in this space include test case optimisation techniques, testing process redesign for software testing and script-less automation testing.

Internet enhances capabilities

Internet capabilities have enhanced instrumentation capabilities in several areas. "LXI devices make remote data acquisition applications relatively simple, synchronisation between tests can be accomplished over the Ethernet using features like IEEE 1588, and remote diagnosis of problems helps minimise downtime," lists Stasonis.

Dr Agrawal explains further that LXI is a recent technological standard that allows the user to plug-and-play test devices from different manufacturers and route the data on a local-area Ethernet network without worrying about the individual system unit's bus structure, operating system or firmware. LXI-compliant equipment also allows a user to place test instrumentation close to the unit under test, thereby saving time and expense by not needing to lay out long wires to connect distant measurement instruments.

In short, a networking technology like LXI makes it possible to logically string together physically separated T&M instruments for remote data acquisition, processing, etc. But for remote test equipment to work together, time synchronisation is very important. That's because in critical spaces like